Lecture 1: Why Python?

Dr. Aafaq · B.Tech. CSE, 5th Semester · Compact notes (Lutz, Ch. 1)

1 Why Do People Use Python?

Reason	What it means	Reason	What it means
Software quality	readable, consistent, easy to maintain and reuse	Support libraries	"batteries included" + 600,000+ PyPI packages
Productivity	code is 1/3 to 1/5 the size of C/C++/Java; no compile step	Integration	talks to C/C++, Java, .NET, web, databases
Portability	same code on Windows, Linux, macOS, phones, boards	Enjoyment	small, clean, fun to write

Same task, two languages: average of four marks.

```c
#include <stdio.h>
int main(void) {
  int m[] = {78, 92, 65, 88}, s = 0;
  for (int i = 0; i < 4; i++) s += m[i];
  printf("%.2f\n", s / 4.0);  /* 80.75 */
  return 0;
}
```

```python
marks = [78, 92, 65, 88]
print(sum(marks) / len(marks))  # 80.75
```

No types, braces, `main` or compile step: two lines say *what*, not *how*.

Predict the output:

```python
b = "engineering"
print(len(b), b.upper())   # 11 ENGINEERING
print(b[::-1], b.count("e"))
                           # gnireenigne 3
```

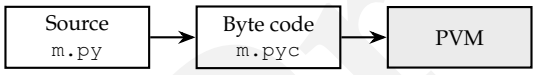
The Zen of Python (`import this`): *Beautiful is better than ugly. Explicit is better than implicit. Simple is better than complex. Readability counts. There should be one obvious way to do it.*

2 Is Python a "Scripting Language"?

It depends on what "scripting" means. Python is used for **shell tools** (automating OS tasks), as a **control language** (glue that drives components written in C/C++), and for **ease of use** (quick, flexible coding). But it is a full **general-purpose** language used to build large systems, so "scripting" undersells it.

3 The Downside: Execution Speed

Python compiles source to **byte code**, which the **Python Virtual Machine** (PVM) runs one instruction at a time. Byte code is not machine code, so plain Python loops are slower than C.




```python
total = 0                          # A: ~0.9 s
for k in range(10_000_000):
    total += k
total = sum(range(10_000_000))     # B: ~0.1 s
```

B is about 9× faster because `sum` runs in C. Heavy work goes to C-based libraries (NumPy, pandas, PyTorch); timings vary by machine.

Key idea. Developer time costs more than CPU time. Write it in Python; push the hot spots into C-backed tools.

4 Who Uses Python, and for What?

Organisation	Use	Organisation	Use
Google, YouTube	infrastructure, tooling	NASA, CERN	scientific computing
Instagram, Spotify	web back ends, data	ISRO, IITs	research, teaching
OpenAI, Meta, DeepMind	AI (PyTorch, TensorFlow)	Netflix, Dropbox	automation, services

What you can do: systems programming (files, processes, sockets); GUIs (tkinter, Qt); web (Django, Flask, FastAPI); component integration (C/C++ via ctypes, Cython); databases (`sqlite3`, SQLAlchemy); rapid prototyping; numeric and scientific work (NumPy, SciPy, pandas, Matplotlib); AI and data science (scikit-learn, PyTorch); games, robotics and embedded boards (pygame, ROS, MicroPython).

Support: open source, managed by the **Python Software Foundation** (PSF); changes are proposed as **PEPs** (e.g. PEP 8 style guide). Guido van Rossum created it (1991) and was "BDFL" until 2018; since 2019 an elected **Steering Council** decides.

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5 Technical Strengths

- **Object-oriented**, but OOP is optional.
- **Free** and open source.
- **Portable**: byte code runs wherever a PVM exists.
- **Powerful**: dynamic typing, automatic memory management, built-in types and tools, libraries.
- **Mixable**: combine with C, C++, Java.
- **Easy to use and learn**: type and run.

```
x = 42; x = "forty-two"   # dynamic typing
marks = [65, 92, 78]      # built-in list
marks.append(88); marks.sort()
print(marks)              # [65, 78, 88, 92]
info = {"name": "Asha", "cgpa": 9.1}
print(info["cgpa"])       # 9.1
print(2 ** 100)           # no overflow
print("spam " * 3)        # spam spam spam
```

Name: from the comedy group *Monty Python*, not the snake; hence "spam" and "eggs" in examples.

6 Python vs Other Languages

	Python	**C**	**C++**	**Java**
Typing	dynamic	static	static	static
Run model	byte code + PVM	native machine code	native machine code	byte code + JVM
Memory	automatic	manual	manual / RAII	garbage collected
Code size	shortest	longest	long	long
Speed	slower	fastest	fastest	fast

Key idea. Python is engineering, not art: built for readability, so teams can maintain code for years.

7 Exercises

1. Give six reasons people choose Python.
2. Rewrite the class-average program in C and Python; count the lines.
3. Is Python a scripting language? Justify.
4. What is Python's main drawback, and how is it handled in practice?
5. Name five organisations using Python and what for.
6. What is a PEP? Who governs Python today?
7. Predict: `a = [4, 9, 1]; print(sum(a))`.
8. Why is Python named "Python"?

Answers. 1: quality, productivity, portability, libraries, integration, enjoyment. 2: about 7 lines of C vs 2 of Python. 3: yes for shell tools, control and ease of use, but it is general-purpose. 4: speed; use built-ins and C-based libraries (NumPy). 5: see the table in Section 4. 6: a Python Enhancement Proposal, the design document for a change; the elected Steering Council, under the PSF. 7: `14`. 8: Monty Python's Flying Circus.

Read next: Lutz Ch. 2–3, How Python Runs Programs. Demo: `lecture01_demo.py`.

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